

Global fishmeal and fish-oil supply: inputs, outputs and markets.

Recent data on fishmeal and fish-oil supply are presented identifying key producer countries and raw material sources and distinguishing between whole fish and by-products. The conversion of these raw materials into marine ingredients is discussed and global volumes presented. This is followed by a summary of the main countries using these marine ingredients over recent years. Uses of fishmeal and fish-oil by market segment are then presented. From this, a global mass balance of inputs and outputs is derived which allows the calculation of the input-to-output ratios (fish in:fish out; FIFO) for the main aquaculture production types to be made. Current areas of focus by the industry include the need to demonstrate sustainable practice, more strategic use of marine ingredients, greater use of fishery and land-animal by-products as well as vegetable substitutes, and novel sources of essential omega-3 fats, notably the long-chain polyunsaturated fatty acids, eicosapentaenoic (EPA) and docosahexaenoic (DHA) acids. Implications are drawn for future supply prospects of fishmeal and fish-oil and their future role in aquaculture, agriculture and human health.

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